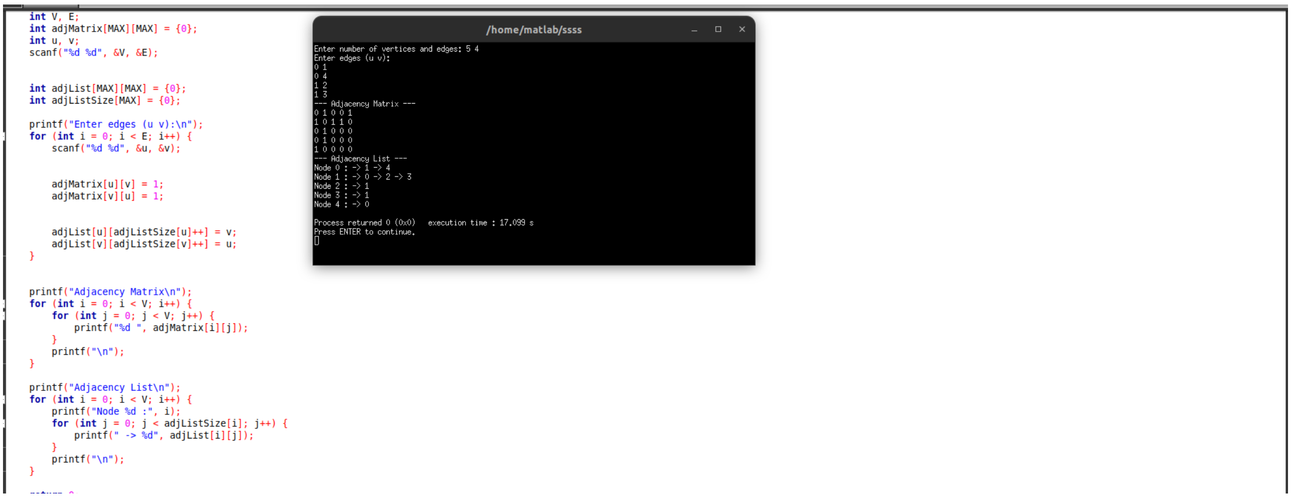
Lab task :1
DAA(LAB)
GRAPHS

NAME:- ARYAN SINGH SIKARWAR
REG NO:- 24BCE0403

que -1 adjacency matrix and list
result-: code completely executed and got the output



que-2 cyclic graph
result:- code completely executed and got the output

part 1(system safe)

```c
#include <stdio.h>
#include <stdlib.h>

#define MAX 100

int dfs(int node, int visited[], int recStack[], int adjMatrix[MAX][MAX], int V) {
    visited[node] = 1;
    recStack[node] = 1;

    for (int i = 0; i < V; i++) {
        if (adjMatrix[node][i] == 1) {
            if (!visited[i] && dfs(i, visited, recStack, adjMatrix, V)) {
                return 1;
            } else if (recStack[i]) {
                return 1;
            }
        }
    }

    recStack[node] = 0;
    return 0;
}

int isCyclic(int adjMatrix[MAX][MAX], int V) {
    int visited[MAX] = {0};
    int recStack[MAX] = {0};

    for (int i = 0; i < V; i++) {
        if (!visited[i]) {
            if (dfs(i, visited, recStack, adjMatrix, V)) {
                return 1;
            }
        }
    }
    return 0;
}

int main() {
    int V, E;
    int adjMatrix[MAX][MAX] = {0};
```

```
/home/matlab/kkkk
 4 3
0 1
1 2
2 3
System Safe

Process returned 0 (0x0)   execution time : 12.487 s
Press ENTER to continue.
```

part 2(deadlock detected)

```c
#include <stdio.h>
#include <stdlib.h>

#define MAX 100

int dfs(int node, int visited[], int recStack[], int adjMatrix[MAX][MAX], int V) {
    visited[node] = 1;
    recStack[node] = 1;

    for (int i = 0; i < V; i++) {
        if (adjMatrix[node][i] == 1) {
            if (!visited[i] && dfs(i, visited, recStack, adjMatrix, V)) {
                return 1;
            } else if (recStack[i]) {
                return 1;
            }
        }
    }

    recStack[node] = 0;
    return 0;
}

int isCyclic(int adjMatrix[MAX][MAX], int V) {
    int visited[MAX] = {0};
    int recStack[MAX] = {0};

    for (int i = 0; i < V; i++) {
        if (!visited[i]) {
            if (dfs(i, visited, recStack, adjMatrix, V)) {
                return 1;
            }
        }
    }
    return 0;
}

int main() {
    int V, E;
    int adjMatrix[MAX][MAX] = {0};
```

```
/home/matlab/kkkk
3 3
0 1
1 2
2 0
Deadlock Detected

Process returned 0 (0x0)   execution time : 22.621 s
Press ENTER to continue.
^[[2~
```